

The Rank the Task Demands: A Causal Rank Law for Matrix Memories Trained on Group Composition

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Abstract

Matrix-valued memories make rank the natural budget of a learned representation: the number of independent directions a state spans bounds what it can bind, compose, and track. We report causal evidence, from two synthetic testbeds trained under a hard single-state bottleneck with an exact-continuous-recovery readout, that gradient descent recruits precisely the rank the task’s algebra demands. On K -pair associative binding, where exact recovery provably requires state rank at least K , trained states develop effective rank tracking K (8.20 at $K=8$, 15.08 at $K=16$ for $d=16$), and train-time rank caps confirm the bound causally: at $d=8$, $K=4$, rank 3 yields zero exact recovery while rank 4 yields 0.94. On group-composition state tracking over five finite groups spanning the solvable/non-solvable divide, the recruited rank equals the group’s minimal faithful real representation dimension $d_{\min}$ (Spearman $\rho = 0.9747$, the design’s tie-capped maximum), the dimension-matched solvable/non-solvable pair S_4/A_5 is statistically equivalent under a pre-registered equivalence test, and a pre-registered force-rank razor is exact in both directions: one rank below $d_{\min}$, recovery is capped by the target’s tied unit spectrum at $\sqrt{(d_{\min}-1)/d_{\min}} \leq 0.894$, below the 0.9 threshold in every group by construction, with observed cells reaching 76–95% (mean 88%) of that forced ceiling; at $d_{\min}$, not geometrically guaranteed to be learnable, recovery clears the anchor-relative bar in all five groups. Representation theory, not computational complexity class, sets what gradient descent buys.

Keywords: effective rank, fast weights, group representations, state tracking

1. Introduction

A matrix-valued memory represents by spanning: whatever a $d \times d$ state binds or tracks, it holds in the geometry of its column space, and rank is the budget it spends. Prior work treats this budget descriptively (Nazari and Rusch, 2026; Sun et al., 2026) or through hand-built constructions (Nichani et al., 2025); neither answers what a geometric account of learned representations needs answered: when a task’s algebra fixes a minimal representational dimension, does gradient descent recruit exactly that dimension, and is it causally load-bearing?

We answer both halves affirmatively under one discipline: a hard single-state bottleneck (the decoder reads only one matrix Z , verified by a gradient blank-out test) and exact-continuous-recovery scoring (cosine against the true continuous target, never argmax over a codebook, which would let a rank-1 state recover on the order of d associations (Nichani et al., 2025)). Section 2 includes the provable K -pair-binding foundation. The headline is group-composition state tracking, where representation theory supplies the ground truth: across five finite groups spanning the solvable/non-solvable divide, the minimal faithful real representation dimension $d_{\min}$ predicts the recruited rank almost perfectly, a matched-dimension solvable/non-solvable pair lands statistically equivalent (Section 3), and a pre-registered force-rank razor is exact in both directions (Section 4): $d_{\min}-1$ is analytically

incapable of exact recovery under the 0.9 threshold ($\sqrt{(d_{\min}-1)/d_{\min}} \leq 0.894$), and $d_{\min}$ suffices, which is not guaranteed a priori. A repaired wave, after an earlier instrument defect nulled the razor (Appendix C), landed this prediction.

2. Tasks, Models, and the Rank Instrument

Tasks. In the binding task, each episode presents K key-value pairs, the encoder writes one state $Z \in \mathbb{R}^{d \times d}$, prediction is the literal unbind Zk_j , and rec@0.9 is the fraction of queries with cosine to the true value above 0.9. The group task is the word problem of a finite group G : a word $w = g_{i_1} \cdots g_{i_L}$, drawn as a random walk on G 's Cayley graph ($L \sim \text{U}\{1, \dots, 8\}$), must map to $\rho_G(g_{i_1} \cdots g_{i_L})$ under a pinned orthogonal reference representation of dimension $d_{\min}(G)$, the minimal faithful real representation dimension. The groups are S_3, S_4, A_5, S_5, A_6 with $d_{\min} = 2, 3, 3, 4, 5$ (Appendix A); the first two are solvable, the rest are not, and non-solvable word problems are NC^1 -complete (Barrington, 1989; Barrington and Thérien, 1988). $d_{\min}$ is a different, representation-theoretic axis; S_4/A_5 , matched at $d_{\min} = 3$ with opposite solvability, are the designed dissociation pair.

Model. A Transformer encoder with d_{state} learned row-reader queries maps each word to a single state $Z(w) \in \mathbb{R}^{d_{\text{state}} \times d_{\text{state}}}$, $d_{\text{state}} = d_{\min} + 2$, leaving spare dimensions so over-recruitment is expressible. The target embeds the reference block-diagonally ($\rho_G(\cdot) \oplus I_2$ in the first sweep, $\rho_G(\cdot) \oplus 0$ in the repaired wave); the loss is cosine distance, the readout is fixed with no learned weights that could launder rank, and the decoder reads only Z (blank-out verified). Per-group step budgets (8k–40k) were pinned by pre-registered convergence bars.

Instrument. The learned state is defined only up to scale and an orthogonal change of basis (a Schur intertwiner): $Z(w) \approx cQ[\rho_G(w) \oplus \cdot]Q^\top$. We estimate the model's own dominant $d_{\min}$ -subspace U from the SVD of the *centered* covariance of $Z(w)$ over held-out words, measure **restricted effective rank** (entropy effective rank of $U^\top Z(w)U$), and score **degauged recovery**: cosine after fitting (c, Q) on a fitting split, evaluated on a disjoint split, reported as rec@0.9 . Centering is load-bearing (Appendix B); rank is invariant under the fitted gauge, so a rank-deficient state cannot be degauged into a full-rank one. For force-rank cells the decisional readout was pinned, before the razor wave ran, to the conservative full- Q Procrustes variant (*crosscheck* rec@0.9).

The provable foundation (binding).

Fact 1 *If $Zk_j = v_j$ exactly for K bindings with linearly independent keys and values, then $ZK_{\text{mat}} = V_{\text{mat}}$, so $K = \text{rank}(V_{\text{mat}}) \leq \text{rank}(Z)$.*

The bound is classical (Kohonen, 1972; Anderson, 1972), holding only under exact continuous recovery. Gradient descent recruits the rank: at $d = 16$ effective rank reaches 8.20 at $K = 8$ and 15.08 at $K = 16$, and the recruited rank is causally necessary: a spectral rank cap at $d = 8$, $K = 4$ leaves exact recovery ≤ 0.0004 for every cap $k \leq 3$ and restores it to 0.940 at $k = 4$, a step exactly at the provable bound; the same operator composes exactly (Z^h , 21 self-applications) in four of five seeds, a composition-stability probe periodicity-equivalent to depth 5 under the single 8-cycle target (Liu et al., 2023) (Appendix E).

3. The Rank Law on Group Composition, Observed

Figure 1 (left) gives the observational leg: per-group mean restricted effective rank lands within 5–11% of $d_{\min}$ at every group, and all 19 seeds sit inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band (means and per-seed values in Appendix B).

Spearman correlation between recruited rank and $d_{\min}$ is $\rho = 0.9747$, the maximum this design can express, since the S_4/A_5 tie caps ρ below 1; under the exact permutation null, $P(\rho \geq 0.8) = 8/120 \approx 6.7\%$ (0.8 is pre-registered, below the next achievable level 0.87), so by pre-registration this leg is corroborating rather than independently decisive (a disclosed length-robustness split is in Appendix B).

Dimension, not solvability. If recruited rank were set by computational class, the marquee pair should separate; if by representation dimension, coincide. A pre-registered Welch two-one-sided equivalence test on restricted effective rank (margin ± 0.5 rank-units; $n = 5$ per group) observes a difference of $+0.019$ rank-units (se 0.037, df 7.8), with both one-sided tests passing at roughly seven times the critical value ($t = 13.06$ and 14.12 against $t_{\text{crit}} = 1.865$): equivalence is declared decisively. The pair lands together on dimension, where the rank law puts it.

4. The Causal Razor

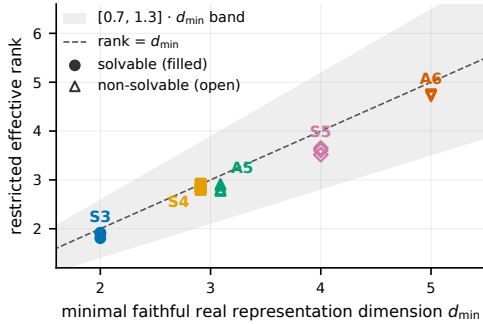
The decisive test intervenes on rank: the encoder’s output is spectrally truncated to rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$ throughout training, against the zero-padded target $\rho_G(\cdot) \oplus 0$ of rank exactly $d_{\min}$, alongside an unconstrained anchor from the same family. Pre-registered reading: if $d_{\min}$ is load-bearing, $k = d_{\min}-1$ must fail and $k \geq d_{\min}$ must recover past $0.9 \times$ the anchor’s crosscheck $\text{rec}@0.9$.

Figure 1 (right) and Figure 2 show the result on the pre-pinned crosscheck readout. Necessity below $d_{\min}$ is an analytically forced floor: against a $d_{\min}$ -tied-unit-singular-value target, the maximum cosine any rank- $(d_{\min}-1)$ operator can reach is $\sqrt{(d_{\min}-1)/d_{\min}} \leq 0.894$, below the 0.9 threshold in every group by construction, so $\text{rec}@0.9 = 0.000$ at $k = d_{\min}-1$ follows regardless of training quality; observed cells still reach 76–95% of that ceiling (mean 88%, per-group breakdown in Appendix B), evidence of near-optimal training under the cap rather than collapse. The zero is noiseless in all five groups and all four S_3 seeds. Sufficiency at $d_{\min}$ carries the causal claim, since no such bound guarantees a rank-exactly- $d_{\min}$ solution is reachable: S_4, A_5, S_5, A_6 clear their bars outright, and S_3 ’s decisive cell, inside the pre-stated ± 0.05 marginality trigger, was extended to four seeds: seed-mean 0.5625 against the fixed 0.495 bar confirms the fifth group (per-seed detail in Appendix B). Both marquee groups confirm at $d_{\min}$; the causal criterion is met on the sufficiency leg.

An earlier 58-cell sweep nulled through an instrument defect (an eye-padded target taxing every capped arm; Appendix C); the zero-padded wave above is the registered fix.

5. Related Work and Limitations

Chughtai et al. (2023) reverse-engineers group multiplication via representation theory; here, rank itself, not an irrep decomposition, is the measured and manipulated quantity. Nazari and Rusch (2026) and Sun et al. (2026) measure effective-rank dynamics in pretrained linear-attention models, observationally, with no task algebra fixing a required dimension; Mishra



G	anchor	$k=d_{\min}-1$	$k=d_{\min}$	$k=d_{\min}+1$	bar
S_3	0.550	0.000	0.450*	0.550	0.495
S_4	0.650	0.000	0.800	0.950	0.585
A_5	0.700	0.000	0.700	0.750	0.630
S_5	0.500	0.000	0.600	0.550	0.450
A_6	0.650	0.000	0.650	0.700	0.585

Figure 1: Left: recruited rank follows representation dimension, not solvability. Each point is one seed’s restricted effective rank vs. its group’s $d_{\min}$ (filled: solvable; open: non-solvable); all 19 seeds are in band, and S_4/A_5 coincide at $d_{\min} = 3$. Right: the causal razor on the repaired rank- $d_{\min}$ target (crosscheck rec@0.9 by force-rank k , one seed per cell, unconstrained anchor, pre-registered $0.9 \times$ anchor bar): exactly 0.000 one rank below $d_{\min}$ everywhere (the forced ceiling is ≤ 0.894), clears the $0.9 \times$ anchor bar at $d_{\min}$ in all five groups (bold). *extended to four seeds under the pre-stated ± 0.05 trigger: seed-mean 0.5625 clears the fixed 0.495 bar.

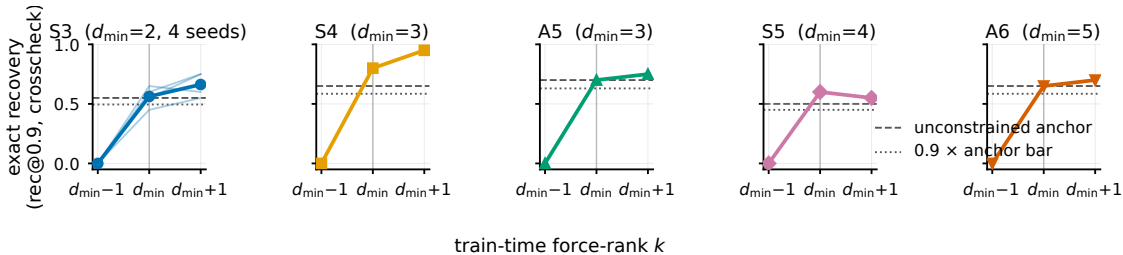


Figure 2: Exact recovery is a step function at $d_{\min}$: per group, crosscheck rec@0.9 on held-out words at force-rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$, vs. the unconstrained anchor (dashed) and the $0.9 \times$ anchor bar (dotted). S_3 overlays its four seeds (thin lines; bold mean); its below- $d_{\min}$ zero is unanimous.

et al. (2026) train matrix-state RNNs on S_3 composition without a rank intervention. [Grazzi et al. \(2025\)](#), [Siems et al. \(2025\)](#), [Merrill et al. \(2024\)](#), and [Delétang et al. \(2023\)](#) characterize which word problems recurrent architectures can express; the marquee equivalence instead sorts by representation dimension. A bolt-on-latent negative result ([Larson, 2026](#)) is this result’s counterpart.

Limitations. All results are on sub-1M-parameter synthetic testbeds, not claims about pretrained language models. Razor cells are single-seed except S_3 (four seeds); S_4/S_5 ’s anchor-bar margins and S_3/S_5 ’s soft convergence at the shortest pinned budget (Appendices B, D) both read directionally, not as an unqualified 5/5.

Appendix A. The Five Groups and Reference Representations

G	$ G $	$d_{\min}$	solvable	realization of ρ_G
S_3	6	2	yes	standard (dihedral)
S_4	24	3	yes	cube rotations
A_5	60	3	no	icosahedral rotations
S_5	120	4	no	standard (zero-sum)
A_6	360	5	no	standard (zero-sum)

Table 1: The five task groups. $d_{\min}$ is the minimal faithful real representation dimension; each reference representation is orthogonal by construction and verified faithful (injective on G) at build time. S_4/A_5 form the marquee pair: matched $d_{\min}$, opposite solvability.

Appendix B. Per-Seed Observational Rank

G	$d_{\min}$	n	restricted eff. rank	band	in band
S_3	2	3	1.877 ± 0.060	[1.4, 2.6]	3/3
S_4	3	5	2.852 ± 0.054	[2.1, 3.9]	5/5
A_5	3	5	2.832 ± 0.062	[2.1, 3.9]	5/5
S_5	4	3	3.591 ± 0.069	[2.8, 5.2]	3/3
A_6	5	3	4.736 ± 0.023	[3.5, 6.5]	3/3

Table 2: Restricted effective rank of the unconstrained trained state (mean $\pm$ sd over seeds) against each group’s $d_{\min}$, with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band. Every seed of every group is in band; Spearman $\rho = 0.9747$ is the tie-capped maximum; the dimension-matched pair S_4/A_5 differs by 0.019 rank-units.

Below- $d_{\min}$ crosscheck cosines as a fraction of the geometrically forced ceiling $\sqrt{(d_{\min}-1)/d_{\min}}$: S_3 86.3%, S_4 91.2%, A_5 94.9%, S_5 75.7%, A_6 93.5% (mean 88.3%). S_5 is the outlier, consistent with its soft-convergence flag (Appendix D).

Per-group deviation of the mean from $d_{\min}$ (Table 2): S_3 6.1%, S_4 4.9%, A_5 5.6%, S_5 10.2%, A_6 5.3%.

S_4 and S_5 ’s razor $k=d_{\min}$ cells (Figure 1, main text) numerically exceed their own unconstrained anchor by 0.10–0.15; plausibly the capped arm skips the ambient dimensions the zero-padded anchor must still learn to null under the same fixed step budget, an unconfirmed hypothesis.

The instrument’s centering step is load-bearing: an uncentered lens scores a flawless synthetic model 0.705 where the centered production lens scores 0.9996 on identical data. The disclosed length-robustness split referenced in Section 3 scores only words of length $L \geq 2$ (the $L = 1$ read is attention-degenerate and was demoted with a mechanism note in the run records): it moves per-group mean recovery cosine by at most 0.013 (per-seed at

most 0.041), with no group-selective divergence. Per-seed, S_3 's $k=d_{\min}$ cell clears its own seed's $0.9\times$ anchor bar in 2 of 4 seeds (seed 1: +0.010; seed 3: +0.110; seed 0: -0.045; seed 2: -0.120); the self-referential bar ($0.9\times$ the seed-mean anchor, 0.574) narrowly fails the seed-mean $k=d_{\min}$ value (0.5625), which is why the main text uses the fixed pre-registered literal (0.495) rather than a self-referential recompute.

Appendix C. The Ambient-Identity Tax in Detail

The first sweep's force-rank target was the eye-padded $\rho_G(\cdot) \oplus I_2$, whose rank is $d_{\text{state}} = d_{\min} + 2$ for every group. Because the identity block is constant across words, it is the cheapest reliable loss reduction available, and a rank-capped arm buys it before the group representation; the residual budget for ρ_G is $k - 2$, below $d_{\min}$ at every grid point, so no capped cell in the original sweep ever tested the law's confirm direction. Three raw-artifact signatures established this: (i) force-rank cells' direct cosine matched the rank- k optimum $\sqrt{k/d_{\text{state}}}$ with mean absolute deviation 0.028 across all 39 cells (maximum 0.166; the two S_5 below- $d_{\min}$ cells are the only outliers, an additional disclosed optimization shortfall), so the arms trained to their rank-constrained ceiling rather than under-training; (ii) centered restricted rank of capped arms landed near $k - 2$, the residual budget; (iii) in the repaired wave, an eye-padded corroboration arm reproduces the failure on demand at raw $k = d_{\min} + 1$ (effective budget $d_{\min} - 1$) while the tax-paid point recovers. The repaired wave's 30 cells were configuration-verified one-by-one against a manifest re-derived independently from the design record (steps, padding mode, and force-rank value per cell; zero skipped steps).

Appendix D. Soft Convergence at the Shortest Pinned Budgets

G	anchor	$k=d_{\min}-1$	$k=d_{\min}$	$k=d_{\min}+1$	bar (0.92)
S_3	0.914	0.665	0.900	0.903	fail (all 4, by margin)
S_5	0.876	0.801	0.879	0.877	fail (all 4)

Table 3: Convergence health (gate1a, min validation cosine over $L \in [2, 5]$) for every S_3/S_5 razor cell at the shortest pinned step budget. All eight cells fall short of the 0.92 margin; S_5 's 0.876–0.879 range is a larger shortfall than S_3 's 0.900–0.914. The necessity leg (Section 4) is unaffected by construction; the sufficiency leg for these two groups should be read as directionally consistent under disclosed soft convergence.

Appendix E. Depth-21 Periodicity Under the Single K -Cycle

The binding task's held-out composition probe (Section 2) uses a single Hamiltonian K -cycle, chosen because a general permutation decomposes into short disjoint cycles that collapse held-out hops into trivial or in-distribution queries. Under a single 8-cycle, $\pi^{21} = \pi^{21 \bmod 8} = \pi^5$ exactly, so nominal depth 21 shares its target with the already-tested depth 5; the two are not distinct group-theoretic targets. What depth 21 measures instead is

numerical stability under 21 sequential self-applications of the trained operator: in the four converged seeds, $\text{rec}@0.9$ at $h = 5$ and $h = 21$ both read ≈ 1.0 , a non-trivial statement that repeated self-application does not amplify error; the fifth (still-transitioning) seed reads 0.303 at $h = 5$ and 0.0001 at $h = 21$, the same underlying imperfection compounding under repetition.

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